



**VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2014
Higher 2**

VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE

CANDIDATE NAME

CLASS

--	--	--	--	--

BIOLOGY

9648/01

Paper 1 Multiple Choice

25 September 2014

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write in a soft pencil.

Do not use any staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

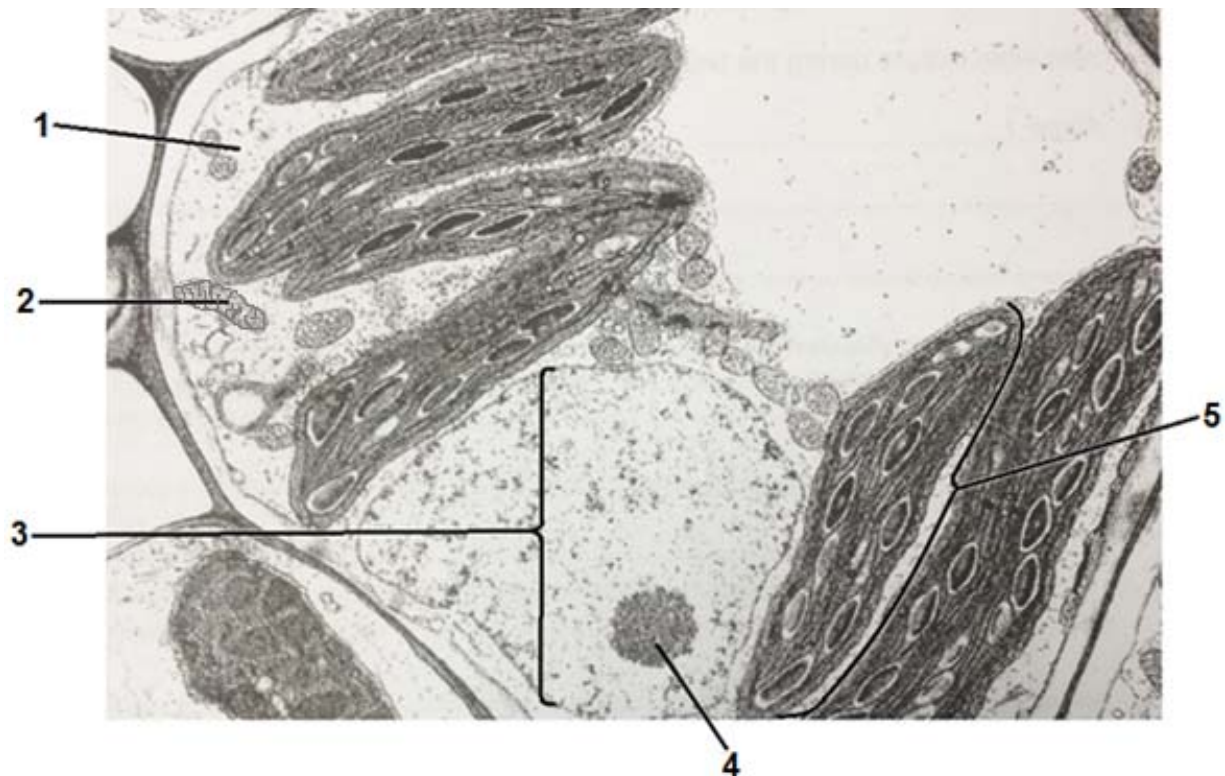
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deduced for a wrong answer.

Any rough working should be done in this booklet

1. The figure below shows an electron micrograph of a plant cell.

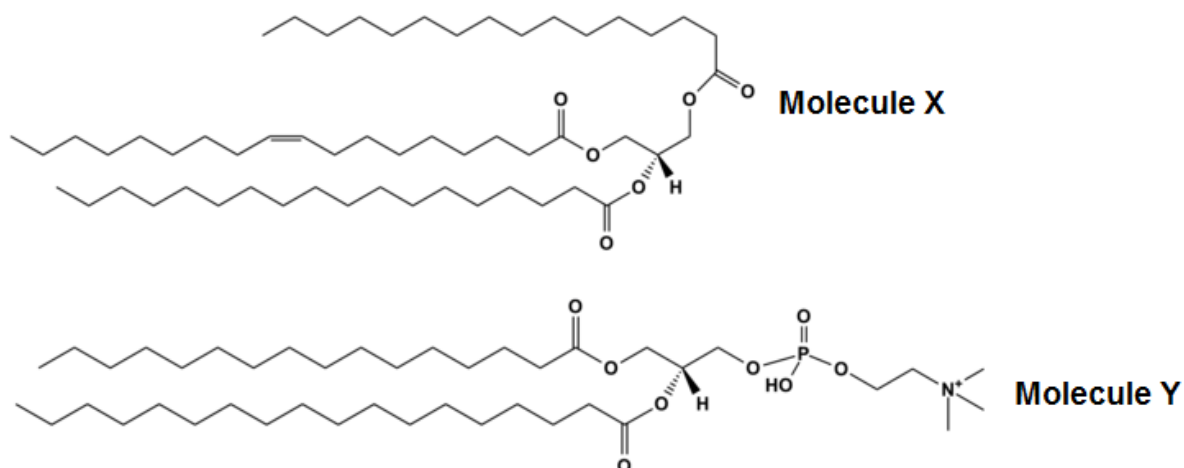


- (i) DNA is found in structures 3 and 5 only.
- (ii) ATP needed to maintain normal cellular functions is supplied by both structures 2 and 5.
- (iii) Ribosomes found in 1 are different from those found in structure 5.
- (iv) The membrane of structure 3 contains pores which allow the passage of both nucleic acids and proteins.
- (v) Structure 4 contains DNA that do not code for proteins.

Which of the above statements are correct?

- A** (i) and (iii)
- B** (ii) and (iv)
- C** (iii), (iv) and (v)
- D** All of the above

2. The diagram shows the structure of two lipid molecules.

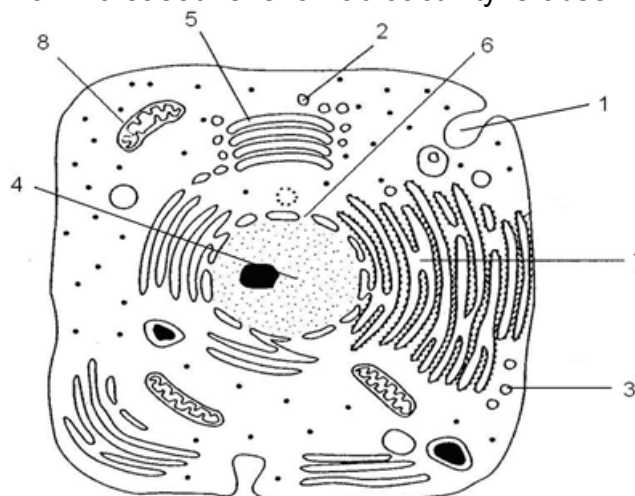


Which statements correctly describe the two lipid molecules?

- (i) Molecule X is amphipathic due to the presence of hydrophobic tails and hydrophilic oxygen atoms.
- (ii) Molecule Y can be used to form vectors for gene therapy.
- (iii) More ATP can be obtained from molecule X than equal mass of glycogen.
- (iv) Membranes in cell contain proteins and molecule Y only.
- (v) Molecule X increases fluidity of membranes as it contains a kink.

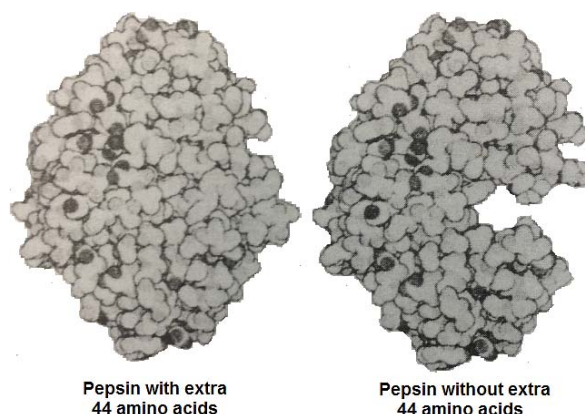
- A** (iii) only
- B** (ii) and (iii)
- C** (i), (ii) and (iv)
- D** (i), (iv) and (v)

3. The diagram shows the ultrastructure of a β -cell isolated from the pancreas. It is grown in a culture containing radioactive amino acids. What is the correct sequence in which increased level of radioactivity is observed?



	First			Last
A	4	6	5	2
B	4	7	3	5
C	7	5	2	8
D	7	5	2	1

4. Protein-digesting enzymes are synthesised inside cells as inactive pro-enzymes, which are then activated outside the cell. Pepsin is synthesised with an extra 44 amino acids. This extra chain is clipped off in the stomach and the enzyme begins to digest proteins.

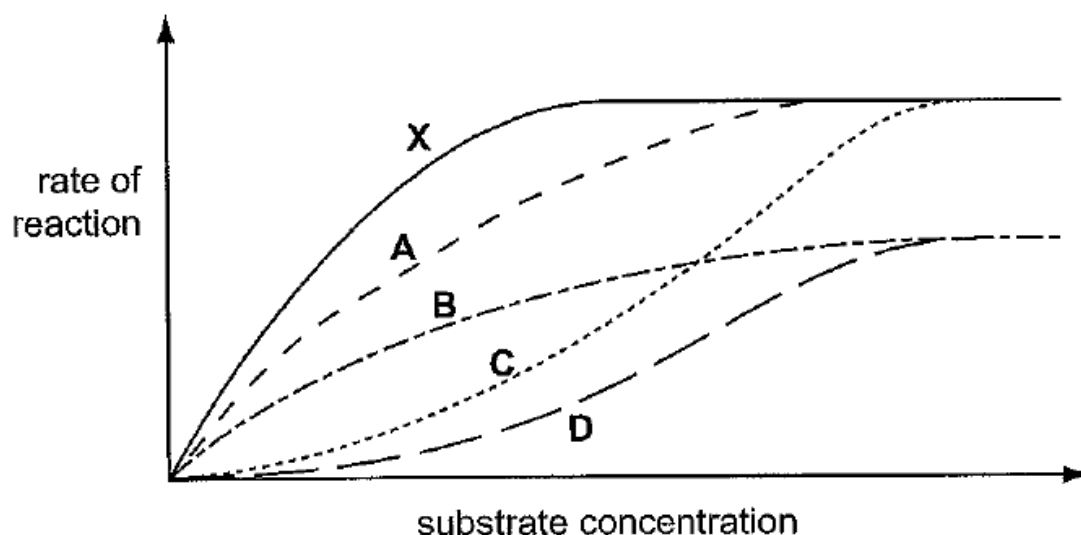


Which of the following is incorrect?

- A There is a change in the tertiary structure of the protein molecule when the extra amino acids are removed.
- B Removal of the 44 amino acids is an hydrolysis reaction.
- C The extra 44 amino acids serve as competitive inhibitor, preventing the binding of the substrate to the active site of the enzyme.
- D The change in shape of the pepsin upon activation is due to a change in the primary structure.

5. In the graph, X represents the initial rate of reaction of an enzyme in increasing substrate concentrations under optimum temperature and pH, in the absence of an inhibitor.

Which curve represents the result when the same experiment is carried out under a lower temperature?



6. Which of the following lists of processes matches the type of movement across membrane correctly?

	Passive diffusion	Facilitated diffusion	Active transport	Endocytosis	Exocytosis
A	Exit of CO ₂ from mitochondria	Chemiosmosis of H ⁺ through ATP synthase	Exit of Ca ²⁺ from synaptic knob	Entry of HIV	Budding of influenza virus
B	Exit of CO ₂ from mitochondria	Exit of K ⁺ from neuron	Entry of Na ⁺ into neuron	Entry of HIV	Budding of influenza virus
C	Entry of O ₂ into mitochondria	Exit of K ⁺ from neuron	Exit of Ca ²⁺ from synaptic knob	Entry of influenza virus	Release of insulin from β cells
D	Entry of O ₂ into mitochondria	Chemiosmosis of H ⁺ through ATP synthase	Entry of Na ⁺ into neuron	Entry of influenza virus	Release of insulin from β cells

7. The figure below shows some *Allium sp.* plant cells in various stages of the mitotic cell cycle, indicated A, B, C and D.

Which of the following correctly describes the structure and / or behaviour of chromosomes at that particular stage?



Stage A



Stage B



Stage C



Stage D

- A** The chromosomes at Stage A are less coiled up as compared to those at Stage C.
 - B** Each homologue has moved to opposite poles of the cell at Stage B.
 - C** The homologous chromosomes are paired up to form bivalents at Stage C.
 - D** The number of chromosomes per nucleus at Stage D is different from that at Stage A.
8. With reference to the figure in Q7, which of the following is false about the structure and / or behaviour of the nuclear envelope and centrioles at the particular stages in the plant cell?
- A** The nuclear envelope is most likely intact at stage A.
 - B** The nuclear envelope has disintegrated at Stage B.
 - C** The nuclear envelope has disintegrated and the centrioles have migrated to opposite poles of the cell at Stage C.
 - D** The nuclear envelopes are most likely intact at stage D and the centrioles are absent.

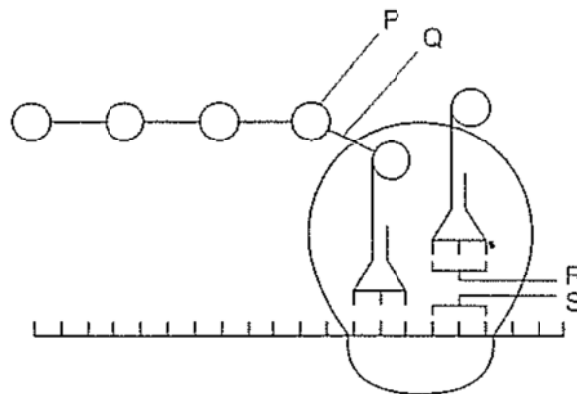
9. Which of the following explains the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?
- (i) The anti-parallel arrangement of the DNA strands.
 - (ii) The RNA primers are required to initiate DNA elongation.
 - (iii) Helicase does not unwind the DNA molecule from the ends but starts at regions in the middle of the molecule to create a replication bubble.
 - (iv) DNA polymerase joins new nucleotides to the 3' end of the growing strand.
- A i and iii
 B i and iv
 C ii and iii
 D ii and iv

10. Transcription in eukaryotic cells results in the formation of pre-mRNA, which is made up of exons and introns.

Which of the following statements correctly describes what happens during the formation of mature mRNA from the pre-mRNA?

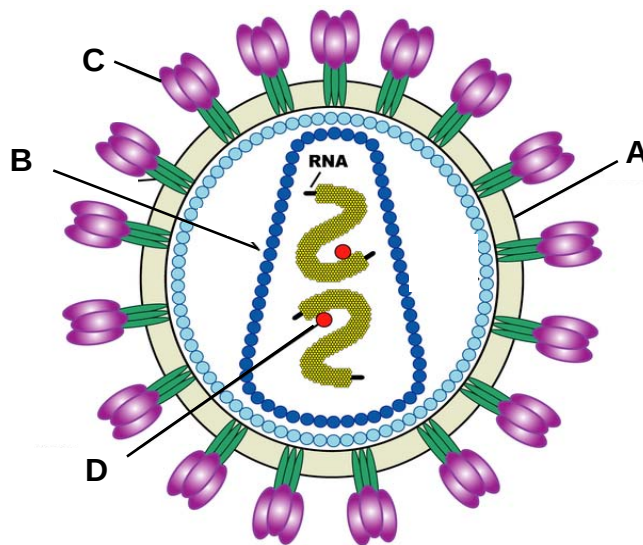
- A All introns are recognised as they share highly similar sequences and are excised.
 B In alternative splicing, some exons may be removed during splicing.
 C All the introns are excised and some of the exons joined together so that they can be transcribed.
 D Alternative splicing takes place, where some of the introns and exons are removed, so that the remaining exons and introns are joined together.
11. The diagram below shows the process of translation in a prokaryotic cell.

Which correctly identifies P, Q, R and S?



	P	Q	R	S
A	N-formylmethionine	peptide bond	anticodon	codon
B	glutamic acid	hydrogen bond	codon	anticodon
C	valine	peptide bond	anticodon	codon
D	ribose	hydrogen bond	codon	anticodon

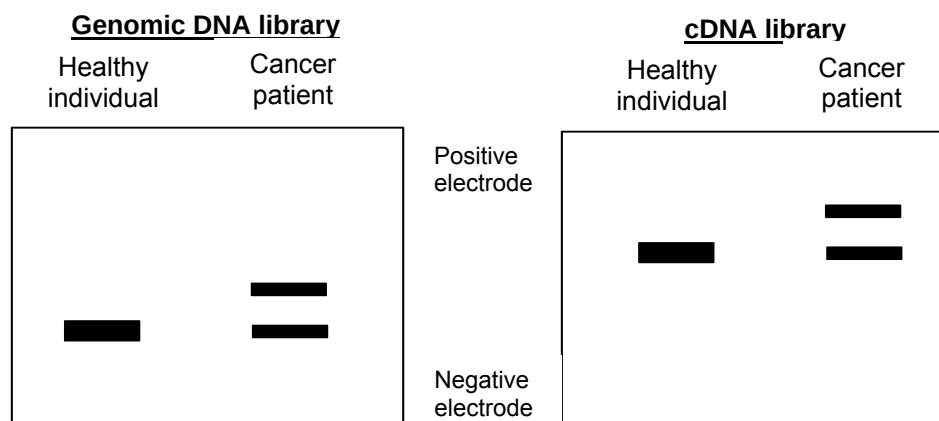
12. To fight influenza viral infections, which of the following strategy is valid?
- A Inhibit the attachment and thereby entry of the virus into its host cell by developing antibodies that bind to the receptors on the viruses.
 - B Inhibit the replication of viral genome by developing RNA nucleotide without 3'OH groups which can be introduced into the infected cells.
 - C Inhibit the expression of viral genes by developing RNA primers which are identical to critical sections of the viral genomes.
 - D Develop antibodies to neuraminidase to prevent entry into the host cells.
13. Which structure does not distinguish the HIV from the T4 bacteriophage?



14. When the products of the lac operon are not made regardless of the presence or absence of lactose. Which of the following statement(s) is/are possible reason(s)?
- (i) Mutation of the *lacI* regulatory gene resulting in a hyperactive protein.
 - (ii) Mutation of operator site.
 - (iii) Mutation of the RNA polymerase binding site
 - (iv) Insufficient amino acids in the bacteria.
- A (i)
 - B (i) and (ii)
 - C (i), (iii) and (iv)
 - D (ii), (iii) and (iv)

15. Which of the following statement does not explain the difference between the trp operon and the lac operon?
- A The structural genes of the trp operon are normally switched on.
 - B The repressor protein of the trp operon is normally inactive.
 - C The lac operon produces polycistronic mRNA.
 - D The lac operon exhibits positive and negative control of gene expression.
16. Which of the following statement regarding the end-replication problem of DNA is incorrect?
- A The end-replication problem will still occur in the presence of telomerase.
 - B The end-replication problem will always occur during the replication of eukaryotic DNA.
 - C The DNA polymerase is the main reason for the end-replication problem.
 - D For every replication cycle, a small section of the 5' end of the original linear DNA is not replicated.
17. Use the following information given below to answer Questions 17 and 18.

The genomic DNA and cDNA libraries of a cancer patient and a healthy individual were created to study the biology of liver cancer. A gene involved in the development of cancer was extracted from both libraries and studied.



Base on the results, which kind of gene was studied and what is the size of the mutant allele with respect to the normal allele?

	Gene studied	Size of mutant allele relative to normal allele
A	Proto-oncogene / oncogene	Mutant allele > normal allele
B	Proto-oncogene / oncogene	Mutant allele < normal allele
C	Tumour suppressor gene	Mutant allele > normal allele
D	Tumour suppressor gene	Mutant allele < normal allele

18. What is the possible cause of the cancer?

- A Over-expression of the gene related to cancer
- B Mutation of gene resulting in a hyperactive protein
- C Additional DNA nucleotides inserted within the mutant allele
- D Gene amplification

19. Some events that take place during the development of cancer are listed below.

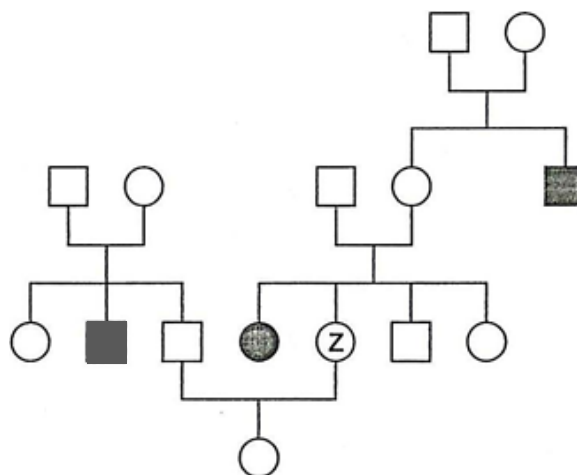
- (i) Accumulations of gene mutations
- (ii) Loss of cell apoptosis
- (iii) Loss of function mutation of p53 gene copy
- (iv) Angiogenesis
- (v) Metastasis
- (vi) Inheritance of a defective p53 gene copy
- (vii) Loss of contact inhibition

In which sequence do these events take place?

- A (iii) → (vi) → (i) → (ii) → (v) → (iv) → (vii)
- B (iii) → (vi) → (i) → (ii) → (vii) → (iv) → (v)
- C (vi) → (ii) → (i) → (iii) → (iv) → (v) → (vii)
- D (vi) → (iii) → (ii) → (i) → (vii) → (iv) → (v)

20. The pedigree below shows the inheritance pattern of a genetic disease that runs in the family of individual Z.

What is the probability that Z will produce a son who is affected?



- A 0
- B 1/8
- C 1/16
- D 1/18

21. A haemophiliac man has one son who has inherited haemophilia and another who has not.

The man's wife is pregnant again. If this baby is a girl, what is the chance that she will have haemophilia?

- A 0%
- B 25%
- C 50%
- D 75%

22. In the Japanese Morning Glory plant (*Pharbitis nil*), flowers can be red, blue or purple. The genotypes which give each of the three flower colours are shown below:

Purple: A₋bb or aaB₋

Blue: A₋B₋

Red: aabb

Which cross would give 20 red, 60 blue and 80 purple in the offspring?

- A AaBb x AaBb
- B AaBb x aabb
- C AaBB x AABb
- D AaBb x Aabb

23. In fruit flies, one gene controls wing form (normal or vestigial) while another gene controls eye colour (brown or red).

A fly with normal wings and brown eyes is crossed with a fly with vestigial wings and red eyes. All the F₁ have normal wings and brown eyes.

When the F₁ are crossed with each other, the resulting F₂ are:

45 normal wing, brown eye
 17 normal wing, red eye
 16 vestigial wing, brown eye
 5 vestigial wing, red eye
 1 normal wing, orange eye

What is the best explanation of the results of this cross?

- A Autosomal linkage
- B Codominance
- C Gene mutation
- D Sex linkage

24. Seeds from a pure breeding plant were planted in identical pots of soil. Sets of ten pots were kept in different light conditions until their first leaves developed. Throughout the experiment, they were watered regularly and all other conditions were kept constant.

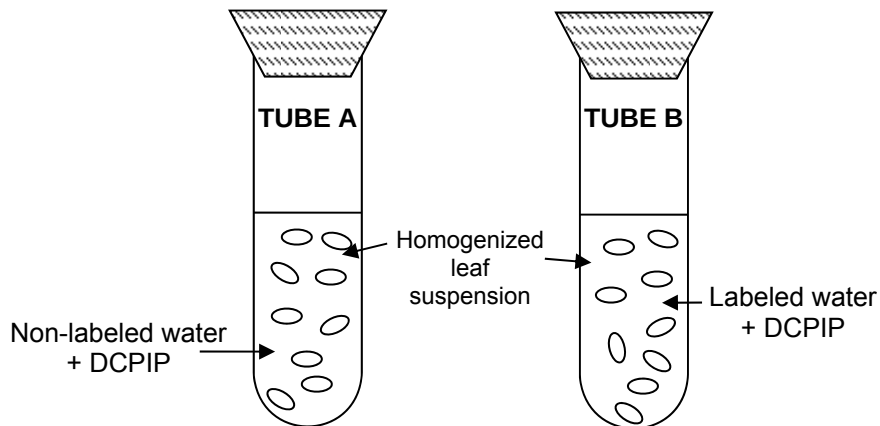
The table below shows the mean height for the young stems, mean length of first leaves and the colour of the leaves.

Which of the following statements explain the effect of light on the phenotype of the young plants?

	No light	Dim Light	Bright light
Mean height / cm	8	6	4
Mean leaf length / cm	1.5	1.4	1.4
Colour of leaves	Pale yellow	Pale green	Dark green

- A** The additive effect of genes is responsible for the continuous variation observed in mean height of young stems.
- B** The length of leaves is most likely to be controlled by polygenes as it shows continuous variation.
- C** The expression of genes involved in chlorophyll synthesis and stem growth are affected by light.
- D** The colour of leaves is likely to be controlled by a single gene as its expression is affected by light.
25. Some photosynthetic organisms contain chloroplasts that lack photosystem II, yet are able to survive. The best way to detect the lack of photosystem II in these organisms would be
- A** to determine if they have inter-thylakoid lamellae in the chloroplasts under a microscope.
- B** to test for liberation of O_2 after exposure for a period of time to light.
- C** to test for CO_2 fixation after 24 hours of darkness.
- D** to test for production of starch using iodine.

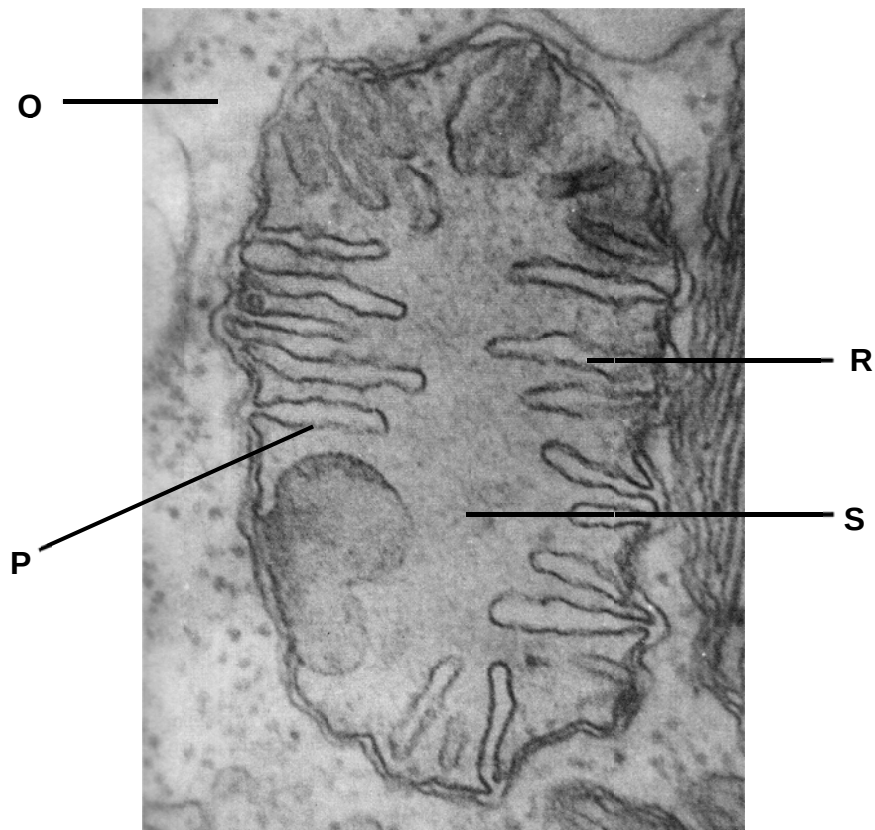
- 26 The experimental setup below was created by homogenizing leaf cells to break their cell walls. The leaf suspensions containing the cytoplasm and organelles were then placed in test-tubes containing non-labeled water (H_2^{16}O) and ^{18}O -labeled water (H_2^{18}O) respectively. A few drops of DCPIP, a hydrogen acceptor, were added to each test-tube. DCPIP will turn from blue to colourless when it is reduced and this colourless DCPIP can be reoxidized to blue.



The test-tubes were then exposed to blue light for 30 minutes. Which of the following shows the results of the two test-tubes after 30 minutes?

	Tube A		Tube B	
	Gas evolved	DCPIP colour	Gas evolved	DCPIP colour
A	$^{16}\text{CO}_2$	Blue	$^{18}\text{CO}_2$	Blue
B	$^{16}\text{O}_2$	Blue	$^{16}\text{O}_2$	Blue
C	$^{16}\text{CO}_2$	Colourless	$^{18}\text{CO}_2$	Colourless
D	$^{16}\text{O}_2$	Colourless	$^{18}\text{O}_2$	Colourless

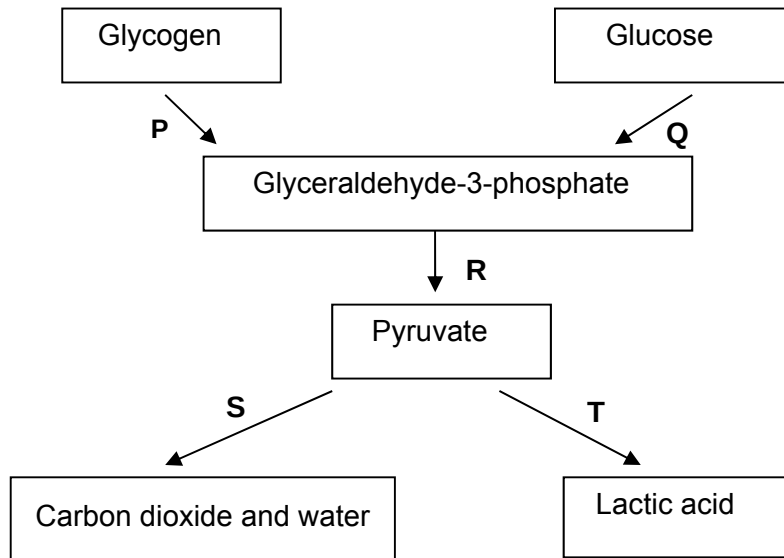
27. The figure below shows an electron micrograph of an organelle.



Match the following processes with the structures labelled **P – S** above.

	Breakdown of fructose-6-phosphate	Oxidative phosphorylation only	Temporary lowering of pH	Formation of reduced co-enzymes
A	Q	R	S	P
B	R	S	P	Q
C	S	P	Q	R
D	Q	P	R	S

- 28 With reference to the diagram below, relate processes P, Q, R, S, T to statements (i), (ii) and (iii).



- (i) NAD is regenerated without the use of the electron transport system
- (ii) ATP is synthesised via substrate level phosphorylation
- (iii) It can take place under anaerobic conditions.

	(i)	(ii)	(iii)
A	R	S	T
B	T	R	Q
C	None of the above	S	R
D	S	Q	None of the above

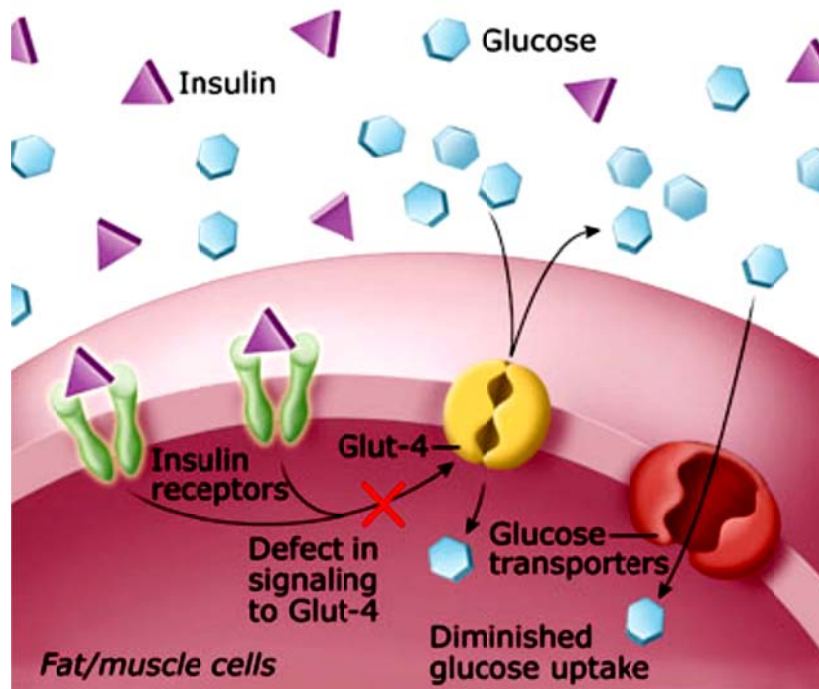
29. The figure below shows the various ions, molecules and transport proteins that are present in a cholinergic synapse.



Which of the following statements correctly describe the transport of substances across membranes as shown in the figure?

- (i) Some substances may diffuse readily across the membrane.
 - (ii) Ions move across the membrane through protein channels via facilitated diffusion.
 - (iii) Some ions are actively transported across the membrane by carrier proteins.
 - (iv) There are at least two transport mechanisms that require energy expenditure.
- A** (ii) and (iii)
B (i), (ii) and (iii)
C (i), (ii) and (iv)
D (i), (ii), (iii) and (iv)
30. Other than non-peptide small molecule neurotransmitters like acetylcholine, the nervous system also utilizes peptide neurotransmitters. In some cases, the same peptide neurotransmitters have been found to have opposing effects on different cells in different locations within the brain.
 Which of the following can be concluded from the statements above?
- A** Alternative splicing produces different peptide neurotransmitters.
B The post-synaptic membrane of one neurone has many neurotransmitter channels.
C The receptors on the post-synaptic membranes of different brain cells have different primary structures.
D The peptide neurotransmitters triggers ligand-gated sodium ion channels in some cell types while they trigger ligand-gated chloride ion channels in others .

31. The following diagrams show the effects of a type of diabetes.



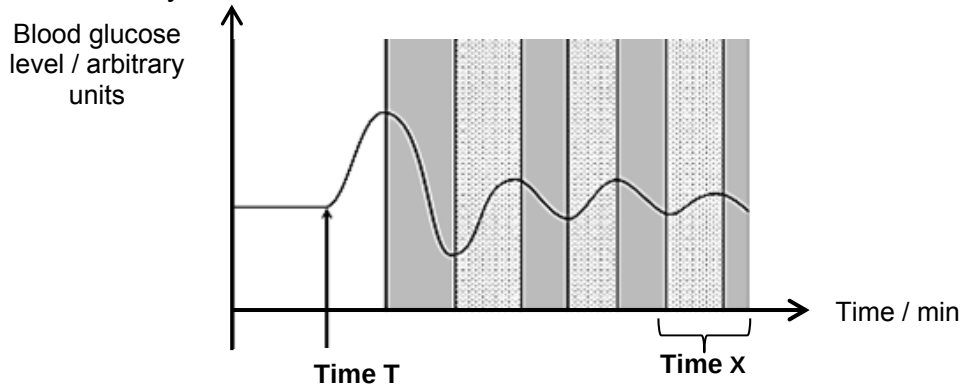
Glucose transporter type 4, also known as GLUT4, is a protein that in humans is encoded by the GLUT4 gene. GLUT4 is the insulin-regulated glucose transporter found primarily in adipose tissues and skeletal and cardiac muscle. Under conditions of low insulin, GLUT4 is embedded within intracellular vesicles.

Which of the following are possible treatments to Type 2 diabetes?

- (i) Introduce recombinant insulin before every meal.
- (ii) Reduce sugar intake.
- (iii) Introduce a mutated insulin receptor gene so that the insulin receptor is activated even in the absence of insulin.
- (iv) Introduce a normal copy of the GLUT4 gene.

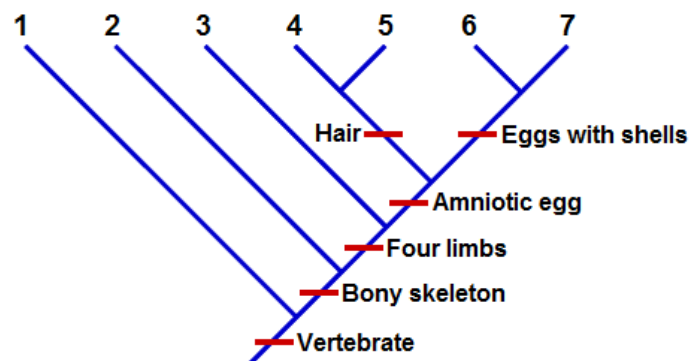
- A** (ii) only
- B** (ii) and (iv)
- C** (iii) and (iv)
- D** (i), (iii) and (iv)

32. The figure shows the blood glucose level of an individual who consumed a meal rich in carbohydrate at time T.



Explain the shape at the end of the graph (time X).

- A Abnormal secretion of glucagon.
 - B Negative feedback loop involving alternative secretion of insulin and glucagon.
 - C Mutation of the insulin receptors on target cells.
 - D Desensitization of the target cells to insulin and glucagon.
33. The following shows a cladogram of 7 different animals.



Which of the following are correct conclusions from the cladogram?

- (i) Animal 3 possess four limbs but not amniotic eggs.
 - (ii) The common ancestor of animal 6 and 7 possessed bony skeleton.
 - (iii) Animal 4 and 7 both possess amniotic egg due to convergent evolution.
 - (iv) For animal 4 and 5, hair is a shared derived character while egg with shell is a shared ancestral character.
 - (v) The evolution of hair would not have been possible without the evolution of amniotic egg.
- A (i) and (ii)
 - B (iii) and (iv)
 - C (i), (ii) and (v)
 - D (ii), (iii) and (iv)

34. Which of the following ideas were **not** part of Darwin's theory of evolution by natural selection?

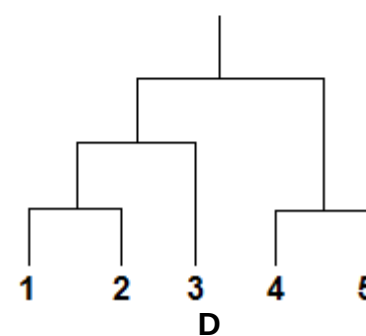
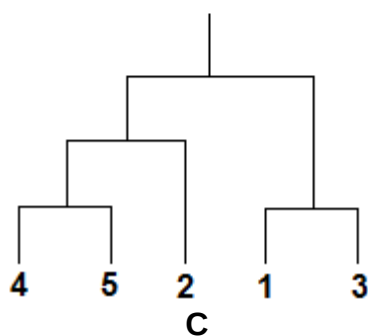
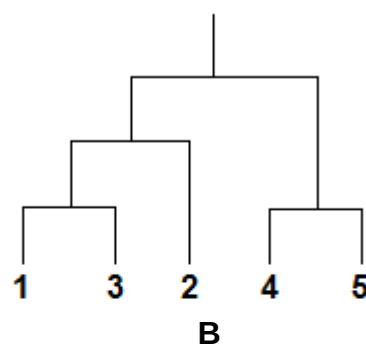
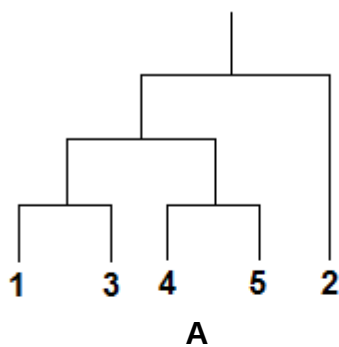
- (i) Acquired traits which increase the fitness of an individual is passed down to the offspring.
- (ii) Organisms tend to produce more offspring than the environment can support.
- (iii) Size of a population keeps increasing as members of the population become more adapted to the environment.
- (iv) Unfavourable alleles will be wiped out from the gene pool of the population. .
- (v) Individuals of the same species in the same population show differences in their phenotypes.

- A (ii) and (v)
- B (i) and (iii)
- C (i), (iii) and (iv)
- D (ii), (iv) and (v)

35. The following shows a section of the homologous gene between 5 different species.

Species 1	A	A	T	T	A	G	C	G	T	A	T	T	A	A	G
Species 2	A	A	T	A	A	T	T	G	T	A	G	T	T	A	G
Species 3	A	T	T	T	A	G	C	G	T	A	T	T	A	A	G
Species 4	T	G	T	A	C	T	C	T	C	A	G	T	A	C	G
Species 5	T	G	T	A	C	T	C	A	C	A	G	T	A	C	G

Which of the following is the correct phylogenetic tree for species 1 to 5?



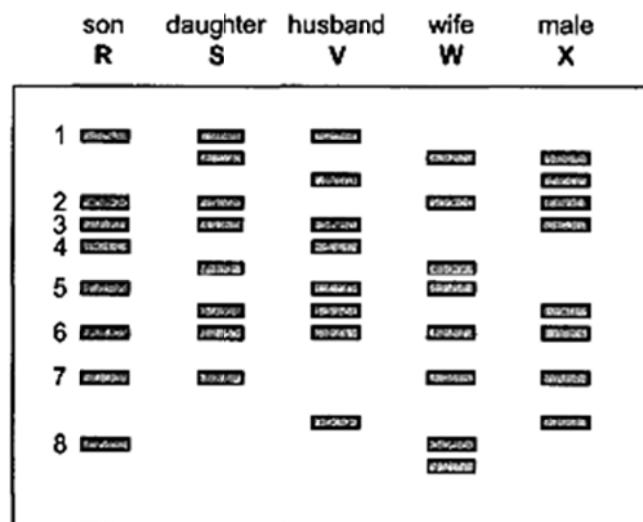
36. Which of the following describes the usefulness of a cDNA library?

- (i) To help isolate homologous genes
- (ii) For expression of eukaryotic genes in prokaryotes
- (iii) To study the expression patterns of genes
- (iv) To determine the effectiveness of anti-cancer drugs.

- A i and iii
- B ii and iv
- C ii, iii and iv
- D i, ii, iii and iv

37. The diagram shows the results of electrophoresis of DNA fragments from four members of a family **R**, **S**, **V**, **W** and another male **X**, who claims that he is actually the father of son **R**. Autoradiography produced the following band patterns.

Which bands indicate that person **V** is the father of son **R**?



- A 1 and 2
- B 1 and 4
- C 2 and 7
- D 3 and 6

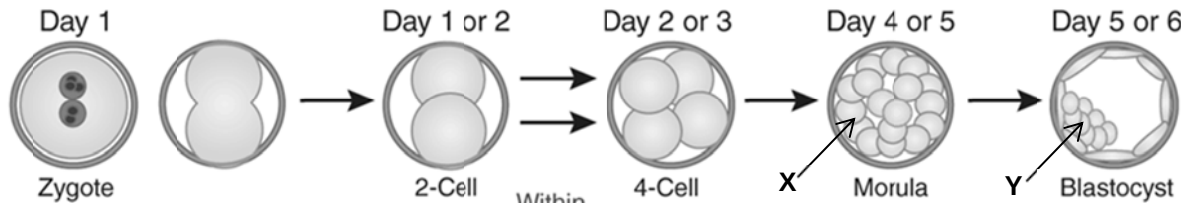
38. At the start of the polymerase chain reaction (PCR), single stranded primers are added to the denatured DNA and the mixture cooled to 60°C.

What explains why the denatured DNA strands anneal with primers and **not** each other at the start of the polymerase chain reaction?

- A The primers anneal only to the 5' end of the denatured DNA
- B The primers are shorter and anneal more easily
- C The primer concentration exceeds the denatured DNA concentration
- D The temperature prevents the denatured DNA from annealing together

39. The figure below shows several stages in the development of an embryo.

Which of the following statements are true about the cells labelled X and Y?



- A X is a pluripotent cell while Y is a multipotent cell.
- B X is a totipotent cell while Y can give rise to multipotent cells.
- C Y will develop into the entire foetus including its placenta.
- D Y will give rise to pluripotent cells but X can only give rise to totipotent cells.

40. Which of the following has the correct information about the genetic diseases and their corresponding forms of gene therapy?

	Genetic Disease	Inheritance Pattern	Gene Therapy
A	Cystic Fibrosis	Sex-linked dominant	Liposomes fuses with plasma membrane of the target cell
B	ADA-SCID	Autosomal recessive	Retrovirus enters target cell via endocytosis
C	X-linked SCID	Sex-linked recessive	Retrovirus envelope fuses with membrane of target cell
D	Cystic Fibrosis	Autosomal recessive	Adenovirus envelope fuses with membrane of target cell

1	C	11	C	21	C	31	A
2	B	12	B	22	D	32	B
3	D	13	B	23	C	33	A
4	C	14	C	24	C	34	C
5	B	15	C	25	B	35	B
6	C	16	D	26	D	36	D
7	A	17	B	27	D	37	B
8	D	18	B	28	B	38	C
9	B	19	D	29	D	39	B
10	B	20	D	30	C	40	C